Gabriel Ong

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Summary / Statement

I have a strong interest in the intersection of Artificial Intelligence and Synthetic Biology, as well as of Biology and Chemistry, and intend to pursue a research career in computational protein design. I hope to be able to work closely with both academia and industry to address real-world problems in sustainability, for instance through novel enzymes for green chemistry. I have rich, holistic research experience in diverse fields, and strong skills in integrating knowledge and concepts across various disciplines to gain novel perspectives for problem solving and discovery. I am also a recipient of the highly competitive and prestigious A*STAR National Science Scholarship, which provides 4 years of PhD funding.

Education

University of Cambridge, Cambridgeshire, United Kingdom
- Undergraduate Student, Natural Sciences (Biological), Sep 2020 to Jun 2024 (expc.)
- Year 1 modules: Biology of Cells, Chemistry, Evolution & Behaviour, Mathematical Biology
  - Mathematical Biology was accompanied by a programming class with R
  - Ranked 39 out of 627 in Year 1 (top 6%)
- Year 2 modules: Biochemistry & Molecular Biology (BMB), Chemistry B (Organic & Inorganic), Pharmacology
  - Each module was accompanied by a laboratory class with at least 60 hours of practical experience
  - Biochemistry skills acquired: Cloning, PCR, Western blot, ELISA, protein purification
  - Chemistry skills acquired: Organic and inorganic synthesis, IR and NMR spectroscopy
  - Pharmacology skills acquired: Tissue drug response assays (guinea pig ileum), antimicrobial assays
  - Ranked 70 out of 575 (Class I) in Year 2 (top 12%). Ranked 3rd & 4th in Biochemistry and Pharmacology respectively
- Year 3 specialisation: Biochemistry
  - Project title: Bioinformatics studies of polyketide synthase dehydratase and enoyl reductase domains
  - Skills: Python, Pymol, protein-protein docking, AlphaFold
  - Ranked 17 out of 44 (Class II.1) in Year 3 (top 38%)
- Year 4 Integrated Masters: Systems Biology
  - Project title: Protein structure networks for variant effect prediction
  - Skills: Python and Machine Learning
  - Ranked 5 out of 20 (Class I) in Year 4 (top 25%)
  - Won Best Research Project award

Hwa Chong Institution, Singapore
- GCE 'A' Level, 7 distinctions, 2012 to 2017
- Editorial team for school science publication Voyager 2016
- Top in Junior College 1 cohort for Chemistry

Academic Employment

A*STAR Institute of Sustainability for Chemicals, Energy and Environment, Singapore
-Research Engineer, Jul 2024 - Jul 2025 (expc.)
-Protein design using protein language models (PLMs)
-Performed structural analysis and molecular dynamics simulations of a PLM-designed unstable RNase A for biomedical applications to better understand mutations suggested by the PLM. Manuscript in preparation
-Also trained models to predict chemical structures from biosynthetic gene cluster sequences, and vice versa, with the aim of expanding the toolkit for retrosynthesis to include enzymes and metabolic pathways for sustainable chemical production. Manuscript in preparation
-Developing in silico screening workflow for PLM-guided enzyme engineering for biomedical and sustainability applications
-Mentored by Dr Winston Koh and Dr Lim Yee Hwee

European Bioinformatics Institute, Cambridgeshire, United Kingdom
-Research Student, Oct 2023 - Jun 2024
-Bioinformatics project on protein structure networks for machine learning-guided variant effect prediction
-Used network-based features derived from network representations of protein structure, in combination with physicochemical properties, to train ML models using deep mutational scanning data for predicting of mutational effects on proteins. Manuscript in preparation
-Gained experience with Python and Machine Learning
-Presented at Frontiers in Science Research Summit at London in 2024
-Won Best Research Project Award
-Mentored by Dr Thodoris Koutsandreas and Dr Evangelia Petsalaki

Department of Biochemistry, University of Cambridge, Cambridgeshire, United Kingdom
-Research Student, Jan 2023 - Mar 2023
-Bioinformatics studies of polyketide synthase dehydratase and enoyl reductase domains
-Used computational protein-protein docking to identify key residues involved in enzyme-substrate and inter-enzyme interactions, with the aim of guiding polyketide synthase engineering efforts for sustainable natural product biosynthesis
-Gained experience with Python, PyMOL, molecular docking, and Linux
-Mentored by Dr Bill Broadhurst

European Bioinformatics Institute, Cambridgeshire, United Kingdom
-Summer Research Student, Jun 2022 - Aug 2022
-Bioinformatics project on enzyme evolution, with emphasis on catalytic residues and mechanism
-Created computational pipeline to infer evolutionary relationships between enzymes, integrating multimodal information on sequence, structure, active site conformation, and reaction mechanisms. This pipeline was adapted to for a study investigating enzyme convergent evolution, which has been published in FEBS Journal.
-Gained experience with Python, PyMOL, Linux, Snakemake, software pipeline development, and various EBI software tools
-Mentored by Prof Dame Janet Thornton

A*STAR Singapore Institute of Food and Biotechnology Innovation (SIFBI), Singapore
-Summer Research Student, Jul 2021 - Sep 2021
-Isolation of bioactive natural product compounds from Fungus and Actinomycetes
-Gained experience with Natural Product Chemistry, HPLC, NMR, Mass Spectrometry
-Mentored by Dr Yoganathan Kanagasundaram

A*STAR Bioinformatics Institute (BII), Singapore
-Research Intern, Feb 2020 - Aug 2020
-Bioinformatics project involving receptor-ligand docking relevant to cancer research and drug design
-Benchmarking of various protein-ligand docking software for rigid and flexible docking using KRAS as a model system
-Gained experience with PyMOL, Python, Linux, and molecular docking software
-Mentored by Dr Srinivasaraghavan Kannan and Dr Chandra Verma

Nanyang Environment and Water Research Institute (NEWRI), Singapore
-Research Intern, Jun 2016 - Dec 2016
-Optimisation of aquaporin-based biomimetic membranes for water treatment through reverse osmosis
-Won Silver Award and Institute of Engineers Special Award at Singapore Science and Engineering Fair 2017
-Won Merit and Best Team Award at Green Wave Environmental Care Competition 2016
-Mentored by Dr Qi Saren and Prof Wang Rong

Publications

Paradigms of convergent evolution in enzymes
Ioannis G. Riziotis, Jenny C. Kafas, **Gabriel Ong**, Neera Borkakoti, Antonio J.M. Ribeiro, Janet M. Thornton (2024)
FEBS Journal; doi: https://doi.org/10.1111/febs.17332

Non-academic Employment

Singapore Armed Forces, Singapore
Automotive Technician (National Service), Jan 2018 - Jan 2020
Corporal First Class (CFC) Rank
Forward Maintenance Section for Exercise Panzerstrike in Germany during Autumn 2019
Worked as part of a team to meet tight deadlines, and adapted to changing requirements in a fast-paced work environemnt

Awards

Winner, Bioinformatics Institute (A*STAR) Poster Session, 2024
Best Research Project Award, Part III Systems Biology, University of Cambridge, 2024
A*STAR National Science Scholarship (BS-PhD), 2019
Silver Award and Special Award (Institute of Engineers), Singapore Science and Engineering Fair, 2017
Merit and Best Team Award, Green Wave Environmental Care Competition, 2016

Academic Service

Cambridge University Synthetic Biology Society

Webmaster, 2021-2022

Non-academic Service

Cambridge University Travel Society

Secretary, 2023-2024
Webmaster, 2022-2023

Churchill May Ball 2022 Committee

Webmaster

Language

Fluent in English. Working proficiency Mandarin Chinese. Basic French and German.

May 8, 2025